

Lab experience 7 Write Up: Lung Sound Analysis with LabVIEW

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Deliverables

- Your lung sound simulation subVI.
- Your in-lab lung sound analysis VI.
- A short write-up containing the following elements:
 - Your name(s)
 - The spectrogram screenshots for Patients 166, 104, and 206. Include a caption for each screenshot, noting where evidence for adventitious lung sounds is present. Feel free to annotate the screenshots if that makes captioning easier.
 - Table 3 (Respiration rate measurements)

Note: up to three students may co-author VIs and write-ups. Documentation of the VIs is optional. You will be graded on completeness.

Table 3. Respiration rate measurements.

Respiration rate timed by a metronome (bpm)	Respiration rate measured by VI (bpm)	% Error $\frac{ metronome - VI }{metronome} \times 100\%$
12		
24		

Tip: Set the metronome to be twice as fast as the respiration rate. Then, inhale at one tick and exhale at another.

Note: If the above breathing rates are making you feel uncomfortable, feel free to just record a few respiration rates that you are used to. If you are experiencing any discomfort, stop the experiment immediately and let an instructor know.